INTERNSHIP REPORT ON

High-Performance Computational Chemistry in Undergraduate Physical Chemistry: Exercises in Homonuclear Diatomic Molecules

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OVERVIEW – COMPUTATIONAL CHEMISTRY

Computational chemistry is a branch of chemistry that uses the results of theoretical chemistry incorporated into efficient computer programs to calculate the structures and properties of molecules and solids, applying these programs to real chemical problems. Examples of such properties are structure (i.e., the expected positions of the constituent atoms), absolute and relative energies, electronic charge density distributions, dipoles and higher multipole moments, vibrational frequencies, reactivity, or other spectroscopic quantities and cross sections for collision with other particles.

Computable quantities include: -

- Structure
- Chemical properties
- Single – molecule properties
- Total energy of a molecule
- Computational properties that do not correspond to physical observables example - bond order, aromaticity etc.
- Additional non observables unique to modelling usually being ties to some aspect of the computational algorithm. An example is atomic partial charge.

Cost and efficiency are other things to be discussed. It has got many intrinsic values. Computational chemistry can be used to predict properties that might be difficult to measure experimentally. The more sophisticated the computational model, more expensive it is in terms of computational resources. Thus, our goal must be to maximize accuracy while minimizing cost. In case of the hardware used, the three things that matter are – processor speed, memory and storage. The software part includes the code including the programming language used. A code renders a model into a set of instructions that the computer can understand. E.g.: - MM3 Model.

Computational chemistry methods range from very approximate to highly accurate; the latter is usually feasible for small systems only; *Ab initio* methods are based entirely on quantum mechanics and basic physical constants other methods are called empirical or semi-empirical because they use additional empirical parameters. Both these methods involve approximations.

Density Functional Theory (DFT) methods are often considered to be *ab – initio* methods for determining the molecular electronic structure. In DFT, the total energy is expressed in terms of the total one-electron density rather than the wave function. DFT methods can be very accurate for little computational cost. Some methods combine the density functional exchange functional with the Hartree–Fock exchange term and are termed hybrid functional methods.

Till recently, Chemical computations was considered an advanced area of research that require high computer skills, power and resources. But it has recently been used as a familiar tool all over the world by chemists of different fields. The thorough understanding of concepts required in computational chemistry has caused it to be reserved for graduation curriculum. But now efforts are made to introduce it to undergraduate curriculum, thus providing the students an early familiarity with this much useful tool. [2-3]

Applications in real life:

- Drug design
- Molecular modelling
- Protein structure prediction
- Quantum dynamics

LITERATURE REVIEW

Introduction

The given article describes computational exercises that are drafted to be incorporated into an undergraduate chemistry curriculum. The exercises are done in the same way as it is done in high level research. The activity is based on the familiar LCAO-MO theory description of homonuclear diatomic molecules. The activity thus introduces the students to various computational chemistry methods and the analysis of computational results. A survey is conducted at the end of the exercise to assess the students' learning gains. The results showed a gain in computational skills of the students', hence achieving the goal.

Computational chemistry, as seen by most people, is not just an advance research field, but can also be used as an effective tool to increase understanding of theories and their applications in real life. Often when computational chemistry is introduced in undergraduate curriculum, they make use of graphical interfaces which are easier to operate and comprehend. But this deprives the students of any practical computational skills as higher level research are performed on command-line based platforms.

The exercise aims to introduce new learners to computational chemistry as it is performed in higher level of research. They are designed to be made a part of the curriculum in order to strengthen the concepts of the theories learned, while introducing familiarity with advance skills. The primary aims are to introduce students to computational chemistry techniques, the basic use of density functional theory (DFT), and the analysis of calculational results. Since most students lack computational skills, the exercise begins with basic computation.

These exercises have been designed to use DFT and the students are exposed to basic components of a DFT calculation. Students learn how to analyse data from DFT calculations, examining the results of structural optimizations and single-point-energy calculations and deriving observables. These exercises were implemented as an independent preparatory homework assignment followed by two consecutive 75 minute in-class lab sessions. These exercises have been incorporated into the Physical Chemistry: Molecular Structure course at Vassar College annually since 2015, with all 90 students in those sections successfully completing the assignment. In Fall 2021, 14 students were given a brief survey at the completion of the assignment to assess their learning gains, and seven students responded. The survey questions primarily focused on gains in student skills, a

deepened understanding of the underlying theory, and students' attitudes toward physical and computational chemistry.

The exercise introduces students to the value of computational experiments as an alternative to wet-lab works. Wrong perceptions of molecular models can hinder learning in introductory courses. Computer visualisations can hence greatly help in solving this issue.

Introduction of computational chemistry courses in undergraduate curriculum

Computational chemistry is becoming more and more important today. It has found its applications in organic synthesis, biochemistry, medicinal chemistry, spectroscopy, or chromatography. "Today, the computer is just as important a tool for chemists as a test tube," the Royal Swedish Academy said in recognizing computational chemists Martin Karplus, Michael Levitt, and Arieh Warshel as the 2013 Nobel Laureates in chemistry. Their efforts have made it possible for scientists to predict and explain chemical processes using computational techniques.[4]

Considering this rising significance of computational chemistry in research field, efforts are being made to incorporate computational chemistry in undergraduate curriculum, as these are usually taught from graduate level. Educating early undergraduate students in computational chemistry is a difficult task. Many times, the students don't have a solid background and even may not require much high pre-requisites to enrol into a chemistry undergraduate course. A proper background in ab initio quantum chemical techniques and computational implementation usually does not come until the graduate school. This is unfortunate as students are unexposed to this important tool in chemical research of today. Efforts in the literature to address this issue have been appearing for over a decade with varying degrees of success. One of the earliest accounts was a review by DeKock, that provides an excellent introduction to the pedagogical issues that arise when teaching quantum theory to students near the beginning of their undergraduate careers. Since then, many have tried to accomplish this task by introducing various strategies and exercises.[5-6]

Apart from being a handy tool for further studies and research, computational chemistry, is helpful to stimulate students to learn chemistry, gain a better understanding of the three-dimensional nature of the science, and learn the value and limitations of computational chemistry techniques as applicable to various subdisciplines of chemistry. It helps students in understanding key concepts such as the molecular orbital theory prepare them for both industrial work and graduate school. Today, the methods have been refined so as to allow even undergraduate students to perform these calculations on proper instructions.[7]

The use of computational molecular modelling to enhance the teaching of chemical concepts is becoming more common too. But this requires some sort of financial investment. This reality poses a problem for institutions where funding and associated resources are scarce and has a potential negative impact on student learning.

The integration of computational chemistry into the curriculum has been difficult due to software and hardware concerns. It has been facilitated due to the availability of more student-friendly software and decreased cost of computing resources.[8]

Many of the institutions use commercially available programs. Inexpensive or free programs such as GAMESS can be used at institutions lacking funds for computational software, along with free front-end interface software, such as WebMO20 or ChemCompute,21 that simplifies building and running jobs for the user.[9]

AN EFFORT IN UNIVERSITY OF WISCONSIN-MADISON

The general chemistry laboratory director, has worked with computational chemistry leader, and, undergraduate research coordinator, to design new lab course that include computational chemistry components. Students are mainly taught to visualise 3D models of molecules. They also learn to connect these models to key concepts of chemical structure and reactivity. An effort to include computational chemistry in the organic chemistry laboratory has also been done. The students were asked to predict and analyse experimental results using computational tools.

Computational chemistry help students explore abstract concepts such as molecular orbital theory. The approach allows students to gain a more sophisticated and accurate understanding of molecular structures and bonding.

The new changes required user friendly software and enough resources required for the many numbers of undergraduate chemistry students. They used WebMO graphical interface. The web-based interface is easy to use and provides access to the powerful calculations available through cutting-edge computational software such as Gaussian and MolPro. It was accessible from any computer and can submit multiple jobs at once. It was difficult for the students to use at first but with practice, is able to use it efficiently and it even helped them have more understanding of the molecules, according to a junior researcher,

This is highly recommended in the light of the new guidelines from American Chemical Society (ACS) require graduating seniors to be familiar with computational chemistry in order to receive an ACS-certified bachelor's degree.

Description of the exercise

The exercise described in this article introduces students to computational chemistry as it is in higher level research. The aim is to introduce students to the various techniques used, with emphasis on the basic use of Density Functional Theory (DFT). The exercise uses DFT to validate the predictions made by the molecular orbital (LCAO-MO) theory of homonuclear diatomic molecules. Since molecular orbital (LCAO-MO) theory is something that students are already familiar with and calculations on homoatomic diatomic molecules being relatively simple, it helps greatly to avoid unnecessary complications. It assists students who are introduced to these techniques for the first time.

The reason to choose DFT over other single-determinant methods like Hartree-Fock (HF) is because it is more commonly used. At this introductory level, students are exposed to only basic components of DFT, such as the choice of basis set and exchange correlation functional. Students also learn to analyse data from DFT calculations. They obtain the result of optimisation and single point energy calculations (i.e., calculations of the wave function and energy at a fixed geometry) and derive observables such as ionization energies, bond dissociation energies, and energies of singlet−triplet splitting.

The exercise had an independent preparatory assignment and two consecutive 75 minute in-class lab sessions.

BACKGROUND

Zeroth-order and minimal-basis LCAO-MO theory are used to make predictions about the structure of homonuclear diatomic molecules (bond orders, spin multiplicity etc). Just as LCAO-MO theory constructs a model of molecular orbitals, their relative energy levels, and their occupancies, so do computational methods such as HF, Kohn−Sham DFT, and other correlated wave function methods build a set of molecular orbitals from atom-centred basis sets, calculate molecular orbital energies, and predict occupancies.

DFT calculations on the singlet and triplet states of homonuclear diatomic molecules should reflect the predictions by LCAO-MO theory. It can also correctly predict the existence of bound state molecules. The bond order predictions of LCAO-MO theory are reflected not only in the bond lengths from DFT optimizations but also by explicit calculations of the bond dissociation energy (BDE) of the ground-state molecule. The BDE is defined as the difference between the total energy of the ground-state diatomic molecule and the total energies of the individual isolated constituent atoms.

$$BDE(X_2) = 2E(X) - [E(X_2) + ZPE(X_2)]$$

Where E(X) is the total energy of the neutral isolated X atom in its ground-state multiplicity, $E(X_2)$ is the total energy of the neutral X_2 molecule, and $ZPE(X_2)$ is the zero-point energy of the X_2 molecule.

In this activity, students are also taught to calculate other physical observables such as the ionisation energy (IE)

$$IE = E(X_2^+) - E(X_2)$$

The vertical IE is the one obtained when the molecular cation is in the same geometry as the neutral molecule. The vertical IE of a singlet molecule can also be approximated as the negative of the eigenvalue of the highest occupied molecular orbital (HOMO). Students can therefore derive the vertical IE of a singlet molecule in two ways and compare the accuracy of the HOMO approximation.

THE INTRODUCTORY INSTRUCTIONAL VIDEO

The main aim of this class was to introduce the students to basic computational knowledge. The class focused on introducing various Unix commands. They were taught to log in to the institution's high performance computer cluster using the Unix's shell. They were also taught to navigate through directories. At the end of this class, a fun assignment was given to provide a better understanding of the directory structure.

GAUSSIAN LAB PART-1

This video aims to familiarize students with the preparation of Gaussian 09 input files, job submission for Gaussian 09 calculations via SLURM, job monitoring on SLURM, and analysis of Gaussian 09 output files.

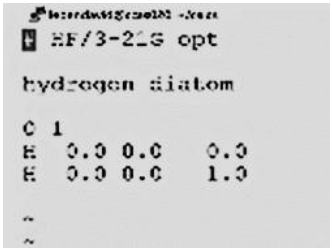
The instructions provided are as follows:

Setting up:

- Log onto Hopper and make sure that you are in the home directory.
- Make directory called "calcs"
- cp/data/pchem/lab1/H2.in calcs/

For Gaussian input file:

- Enter "calcs" directory and open H2.in.
- Overview of H2.in file:



Line 1: Hartree-Fock calculation with 3-21G basis set, optimisation

Line 3: title line

Line 5: charge, spin

Line 6: element x,y,z (coordinates)

Line 7: element x,y,z

Line 8: blank line

H2.in is an optimisation calculation for singlet H_2.

For job submission:

- Ensure that you are in the "calcs" directory.
- Open the G09sub.sh file
- Submit the h2.in calculation to run Gaussian on Hopper.
- Check to make sure job is finished
- New output files in directory.

Analysing output:

- Open output file
- Prints out final energy and equilibrium bond length.

Now, if we need to calculate in triplet state, the spin would be 3 instead of 1 in the input file.

GAUSSIAN LAB PART-2

The students were assigned a unique homonuclear diatomic molecule from the options: He_2, Li_2, B_2, C_2, N_2, O_2, F_2, Ne_2, Mg_2, and Na_2. Students work independently and were first asked to do a LCAO-MO analysis lowest energy singlet and triplet states of their diatomic. They are then asked to use DFT with the B3LYP exchange correlational functional and cc-pVDZ basis set to optimise the structures.

The students then determine the BDE of the singlet state by running a single-point-energy calculation on the isolated neutral atom to obtain its total energy. They compare the prediction of the BDE from DFT with the prediction of the bond strength from the bond order from LCAO-MO theory. Finally, students apply the DFT analogue of Koopmans' theorem to estimate the singlet IE from the HOMO eigenvalue of the neutral singlet diatomic. They then derive the IE by calculating the total energy of the cation doublet diatomic and compare the two methods of predicting the IE.

Reference table of the data obtained by DFT and HF for homonuclear diatomics.

'able S1. Data from DFT (B3LYP/cc-PVTZ) calculations for the full set of homonuclear diatomics used in this activity

	He	Li	B	C	N	O	F	Ne	Mg	Na
a. singlet energy (a.u.)	-5.814	-15.016	-49.382	-75.889	-109.533	-150.271	-199.513	-257.819	-400.183	-324.609
b. singlet bond length (Å)	3.073	2.712	1.652	1.262	1.104	1.210	1.410	2.532	3.929	3.044
c. triplet energy (a.u.)	-4.271	-14.966	-49.416	-75.926	-109.278	-150.334	-199.461	-256.181	-400.118	-324.581
d. triplet bond length (Å)	1.004	2.594	1.628	1.316	1.211	1.209	1.732	1.649	2.776	9.752
e. cation energy (a.u.)	-4.995	-14.821	-49.063	-75.434	-108.959	-149.860	-198.936	-257.129	-399.939	-324.416
f. singlet vertical ionization energy (a.u.)	0.819	0.195	0.320	0.456	0.574	0.411	0.578	0.690	0.244	0.192
g. singlet state highest occupied MO eigenvalue (a.u.)	-0.648	-0.134	-0.215	-0.325	-0.421	-0.244	-0.399	-0.520	-0.179	-0.131
h. zero point energy of ground state molecule	0.0001	0.0008	0.0023	0.0039	0.0056	0.0038	0.0023	0.0003	0.0001	0.0004
i. atomic energy (a.u.)	-2.907	-7.491	-24.661	-37.852	-54.589	-75.068	-99.727	-128.909	-200.091	-162.291
j. ground state bond dissociation energy (a.u.)	0.000	0.032	0.092	0.219	0.350	0.193	0.058	0.000	0.000	0.027

Assessment

The lab had been part of the institution's curriculum for the past 7-8 years and has evolved according to need. It was originally designed as a single in-class session, but in order to allow more time to cover the required material, it was divided into two sessions. Also, live instructions couldn't work as it required all students to work in the same pace, hence it was converted to a video class. Since students found more difficulty learning Unix commands, more time was devoted to explaining directory tree and absolute and relative paths. Earlier, the students were initially taught to use the Nano text editor, but this was changed to vim to help students be more equipped for the more advanced independent research.

In Fall 2021, an online survey was conducted to know the outcome of the activity. It was designed to obtain the self-assessment reports of the students' gains in relevant skills and techniques. Only 7 students participated, limiting the statistical significance of the result.

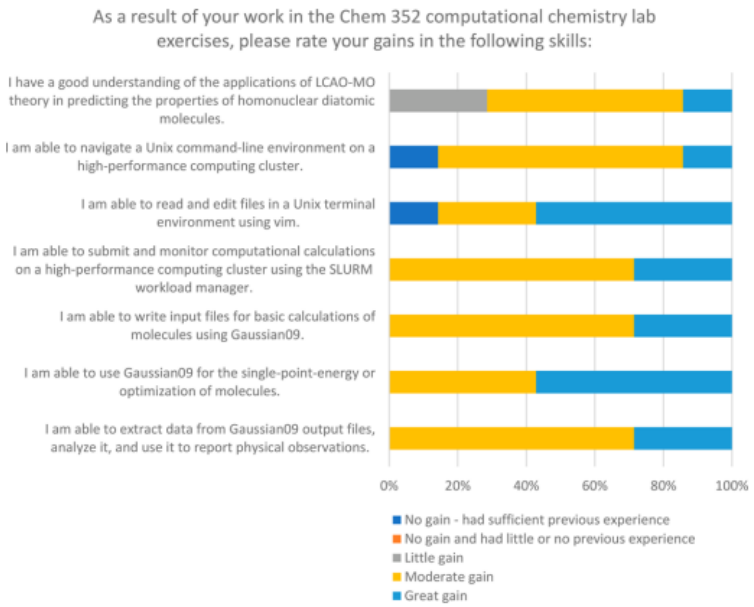


Figure 3. Survey assessing the students' learning gains: questions and student responses from Fall 2021.

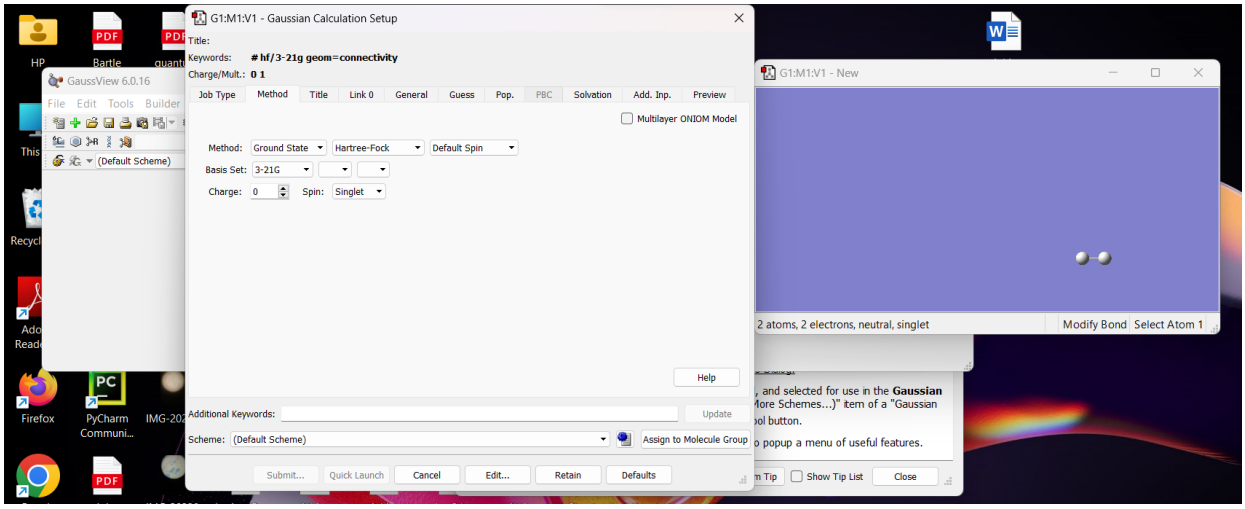
Suggestions and Questions

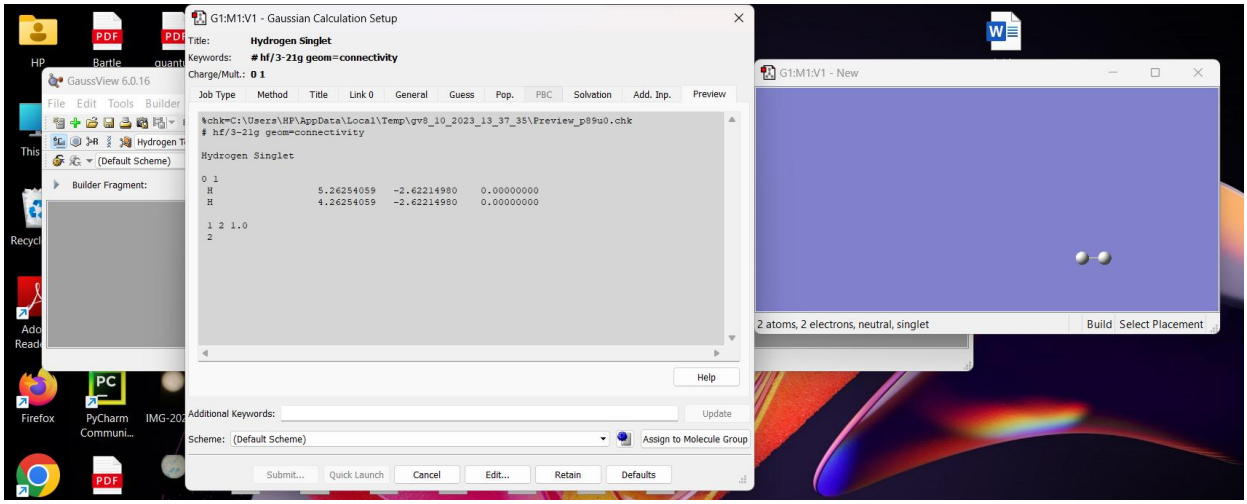
The activity described in this article is with no doubt an efficient way to introduce students to the world of computational chemistry. Since, the exercise aims to equip students for their future high-level research, it more is needed to work with simple homonuclear diatomic molecules alone. Higher level researches deal with more complex molecules: hence the students that participated in this activity may still find it difficult to cope during their higher studies.

Also, students in undergraduate courses usually find it difficult to adjust to the new course structure and environment of learning. Introducing such a new concept at this level of education may also be stressful to many students, as they already have a lot in their hands.

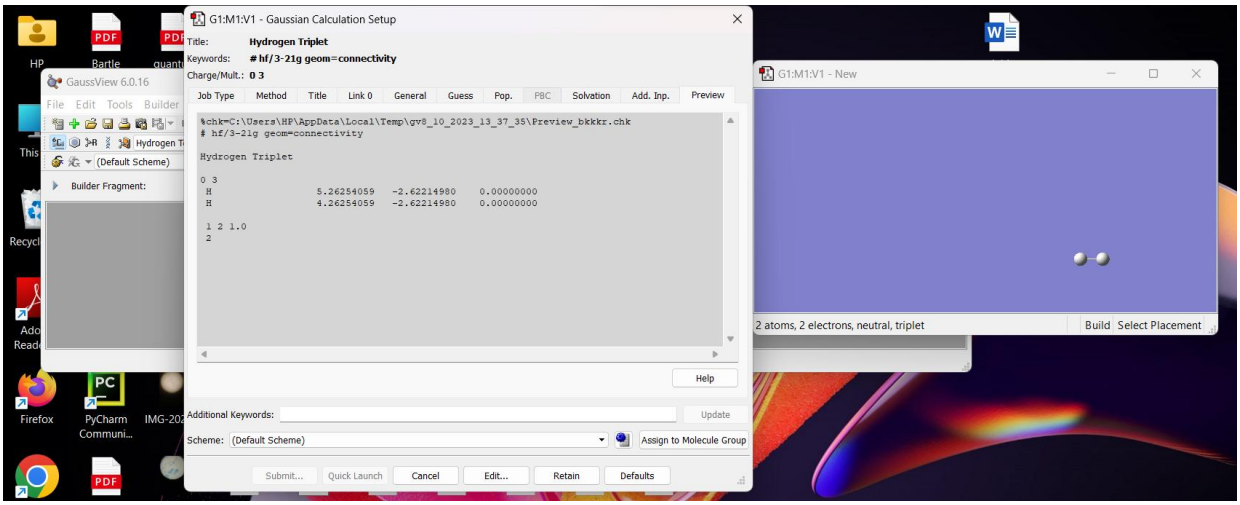
Building molecular structures using GaussView:

1) H_2 molecule, singlet, using Hartree-Fock method

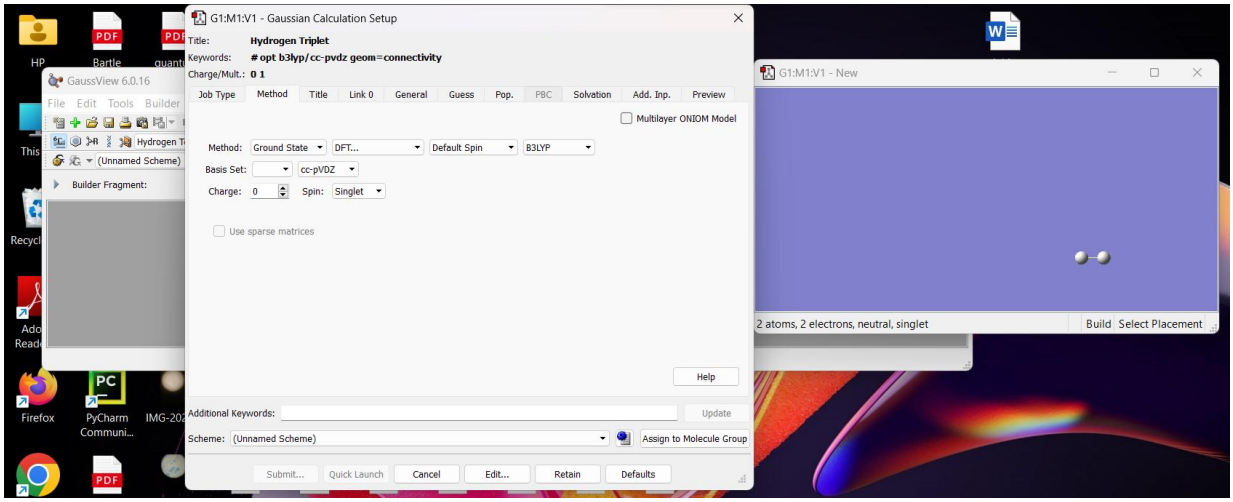


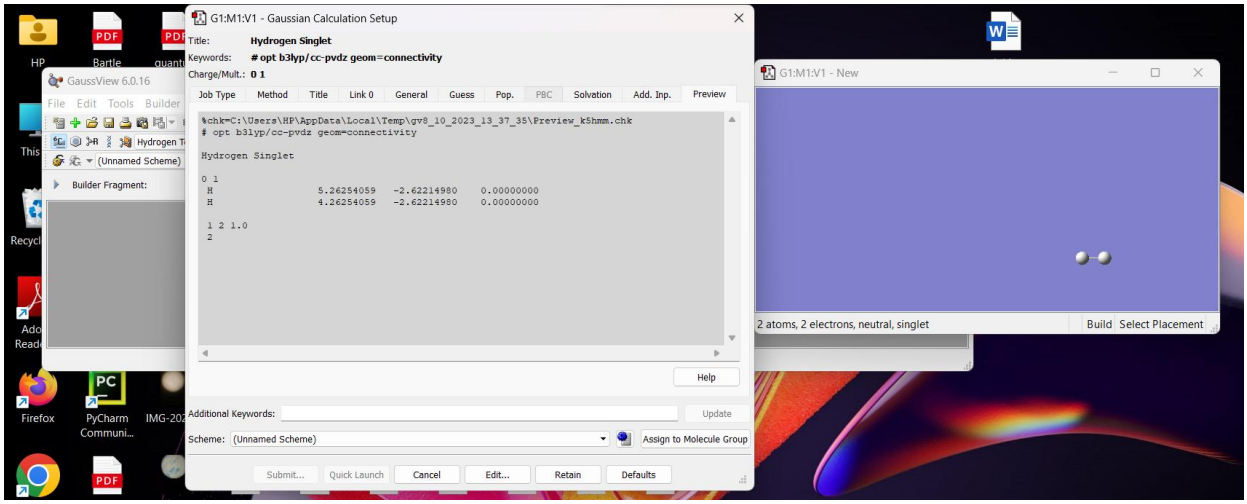


2)H_2 molecule, triplet, Hartree-Fock method



3) H_2 molecule, singlet, using DFT





Conclusion

Overall, this internship has been a wonderful learning experience for me. I was introduced into the field of computational chemistry, in which I had no much knowledge in. I was able to see how students in my level of education could benefit in their studies and conceptual understanding with the help of computational tools. I was prompted to try out some of the calculations using computational methods, within the limits of software and hardware resources available.

I have tried my best to read and refer many texts as to learn more about the given topic. I learned about the various computational methods used and the softwares used like GaussView and Gaussian 16. It has provided me with practical insights into the real world applications of the concepts and theories I have learned.

Moreover, the internship gave me a chance to explore my own strengths and areas for growth. In conclusion, this internship has been a transformative experience that has deepened my understanding of various aspects of chemistry. I am grateful for the opportunity and look forward to applying the knowledge and skills I've acquired to future endeavors.

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